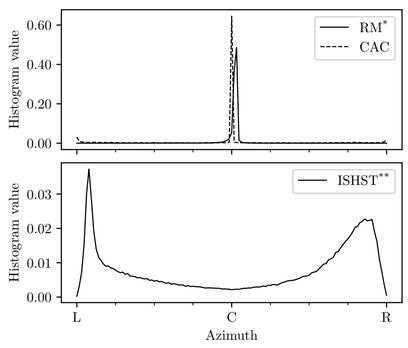
TOWARDS AN OBJECTIVE COMPARISON OF PANNING FEATURE ALGORITHMS FOR UNSUPERVISED LEARNING

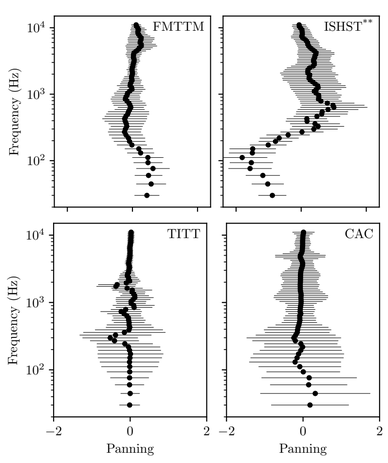
Richard Mitic - richard.mitic@mdu.se ~ Andreas Rossholm - arossholm@spotify.com

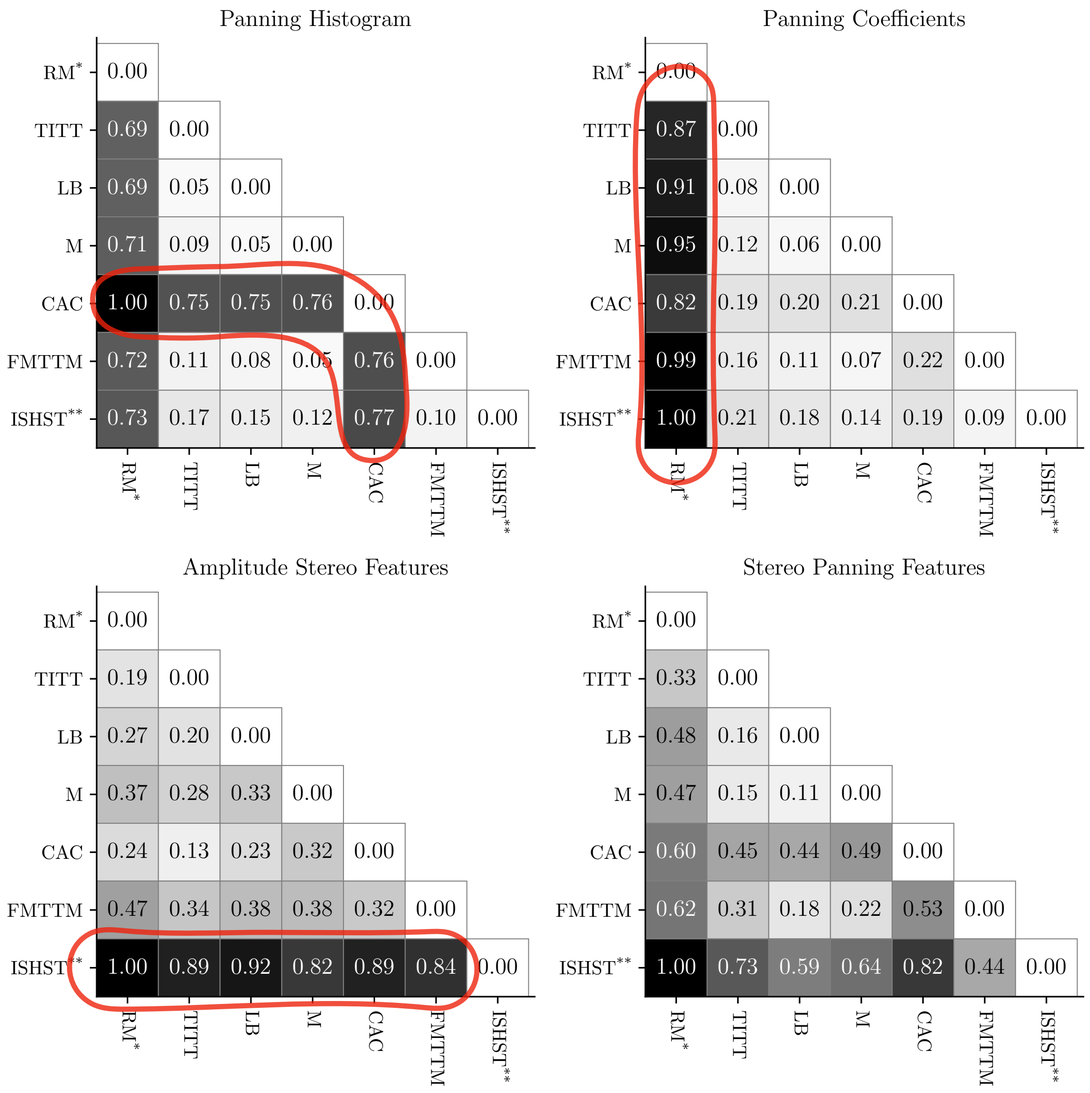
TL;DR - All panning feature algorithms have inherent limitations, but **Stereo Panning Features [1]** currently offers the most promising results for unsupervised learning.

Panning Histograms [2] are **incompatible with Euclidean distance**

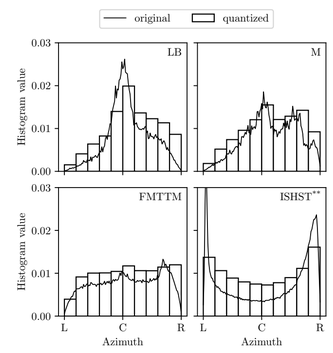


Amplitude Stereo Features [4] place too much emphasis on panning direction, making **most songs appear “very mono”**

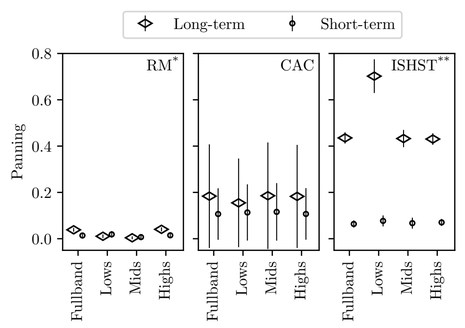




Panning Coefficients [3] cannot represent strong stereo panning, making **most songs appear “very stereo”**



Stereo Panning Features [1] is the **least biased** algorithm with **good separation** between tracks



Problem:
How do you know which panning feature detector to use for unsupervised learning?

Method:
- Extract panning features from a handful of tracks and check the distance between them in euclidean space
- Use anchor tracks that are strongly mono and stereo to constrain the space
- Find errors and biases of each detector

Scan for full paper and supplementary material

References:
1. G. Tzanetakis, R. Jones, and K. McNally, ‘Stereo Panning Features for Classifying Recording Production Style’, in *Proceedings of International Society for Music Information Retrieval Conference*, 2007, pp. 441–444. Accessed: Sept. 12, 2024.
2. A. M. Sarroff and J. Bello, ‘Measurements of spaciousness for stereophonic music’, in *Journal of The Audio Engineering Society*, Oct. 2008.
3. E. Gomez *et al.*, ‘Music similarity systems and methods using descriptors’, US20080300702A1, Dec. 04, 2008 Accessed: Jan. 22, 2025.
4. D. Tardieu, E. Deruty, C. Charbuillet, and G. Peeters, ‘Production Effect: Audio Features For Recording Techniques Description And Decade Prediction’, in *Proceedings of the 14th International Conference on Digital Audio Effects (DAFx-11)*, Paris, FR, Sept. 2011.